

MIT Art, Design & Technology University, Pune

(Established by Govt. of Maharashtra by MIT ADT University Act No. XXXIX of 2015)

MIT SCHOOL OF FOOD TECHNOLOGY, PUNE



Open Elective Courses

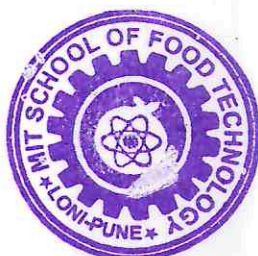
Offered by MIT School of Food Technology

(14th BOARD OF STUDIES IN FOOD TECHNOLOGY)

With effect from Academic Year 2024-25



PRINCIPAL
MIT SCHOOL OF FOOD TECHNOLOGY
LOMI-KALBHOR, PUNE-412201



Open Electives offered by MIT School of Food Technology

Sr. No.	Type of Course	Course Code	Course Name	Credits	Hours / Week			Examination Scheme		
					Lecture	Tutorial	Practical	CA	FE	Total
1	Open Elective	23FT3040	Food Processing	3 (3+0+0)	3	0	0	40	60	100
2	Open Elective	23FT3041	Nutrition and Health	3 (3+0+0)	3	0	0	40	60	100

Open Elective Course for Other Disciplines			
FOOD PROCESSING			
Course Code:	23FT3040	Course Credits:	3
Teaching Hours / Week (L: T: P):	3:0:0	CA Marks:	40
Total Number of Teaching Hours:	45	END-SEM Marks:	60
Focuses on	Skill development	Integrates cross-cutting issues	Sustainability
Course Pre-requisites: 1. Basic understanding of food, processing, preservation concepts such as freezing, heating, drying. 2. Scientific concepts like moisture, emulsion, microorganisms such as yeast, bacteria, mold etc.			
Course Description: a. The course highlights on the basics of food, its function and causes of spoilage which emphasize the need of food processing. b. The major categories of processed foods that are prominently seen in market are included in the course. c. Students develop an understanding about the need for processed foods, and basic processing technology for its production. d. This course provides need base developments in functional foods and their health benefits.			
Course Objectives: This course will enable the students- 1. To understand the basic concept of food, processing, preservation, shelf life. 2. To be aware of the demand and genesis for processing foods. 3. To get fundamental knowledge about various processed foods and their processing technology such as frozen, fermented, sterile foods etc. 4. To learn about the food safety and quality requirements for beverage or thermally processed or dehydrated foods. 5. To understand the concept of functional foods and their health benefits.			
Course Outcomes: On completion of the course, learner will be able to- 1. Explain the fundamental concepts of food, processing, preservation, shelf life. 2. Write down the demand and necessity for processing raw material into processed foods. 3. Analyze any one given problem in food processing and provide solutions in terms of processing parameters or unit operations. 4. Describe the food safety and quality requirements for beverage or thermally processed or dehydrated foods. 5. Suggest appropriate functional food based on the health benefits required by the consumer.			

Unit I	Food Processing and Preservation Methods	7 Hours
Concept of food, functions of food, Genesis for processing food, Causes of food spoilage physical, chemical, Microbiological cause, Concept of shelf life, food preservation and processing, Indian and global scenario of processed food market, Categorization of processed food		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning:	
Unit II	Food Processing	6 Hours
Primary processing- Sorting, grading, washing, cutting, deshelling, Secondary processing- milling, baking, fermentation, Tertiary processing- commercial production of RTE, ready meals, Processed beverages- RTS, Squash, syrup processing, carbonated beverages processing, Extruded snacks-processing		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning:	
Unit III	Food processing equipment	22 Hours
Thermally processed foods: Pasteurized foods- principle of pasteurization, Equipment- Heat exchangers, Commercially sterile foods, Canned foods processing and equipment, Aseptically processed foods- concept and processing operations, Frozen processed foods, Equipment for freezing- Air blast, immersion freezing, cryogenic freezing methods, Effect of freezing on foods. Baked foods: Bread, Cookies and cakes processing, Baking equipment, Effect of baking on food, Fermented Foods- Dahi, probiotic beverages, Fermenter and process of fermentation, Pickled foods, High salt pickles, low salt fermented pickles, Effect of pickling on food, Concentrated foods- principle, equipment, Dehydrated foods- principle, equipment, Case study on applications of dryers and evaporators dairy, spice processing industry, Case study on applications of various heat exchangers, freezers in meat, fish and fruit processing industry		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself:	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning: case study on heat exchanges in beverage industry	

Unit IV	Food safety and quality awareness	5 Hours
Processed food safety and quality Concept of food safety and quality, HACCP concept, principle and application in food industry, TQM tools and techniques, FSSAI standards, Case study on food safety and quality assurance protocol in dairy industry		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself:	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning: Case study on food safety and quality assurance protocol in dairy industry	
Unit V	Food supply chain management	5 Hours
Concept of sustainable food supply chain, local supply chain models for sustainability, Buyer-supplier relationship and agricultural cooperatives, Challenges and limitations for sustainable supply chain, sustainable supply chain implementation case study on any one food industry		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself:	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning: sustainable supply chain implementation case study on any one food industry	
Learning Resources		
Text Books: 1. William C. Frazier, Dennis C. Westhoff, Food microbiology, McGraw Hill Company, 1988. 2. Norman W. Desrosier, The Technology of Food preservation, Medtech; 3rd edition, 2017. 3. P. J. Fellows, Food processing Technology, Woodhead publishing, Third edition, 2015.		
Reference Books: 1. Giridhari Lal, G. S.Siddhappa and G. K. Tondon, Preservation of fruits and vegetable -. Bombay Popular Prakashan, 2009. 2. R. P. Shrivastava and Sanjeev Kumar, Fruit and vegetable preservation, CBS Publishers, Third edition, 2014. 3. Vijaya Khader, Food science and technology, published by ICAR, 2016.		

MOOC/ Video Lectures available at:

ASSESSMENT AND EVALUATION PATTERN

WEIGHTAGE	Continuous Assessment (CA)	End Semester Assessment (ESA)
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CA - Assignments (PBL / Case Study / Presentation / Seminar / Group Discussions / Quiz / Test)

ESA - Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating

Experiential Learning - Internship (Summer / Winter) / Industry Visit / Site Visit / Field Trips

Project Based Learning – Mini (Minor) Project / Major Project

Course Articulation Matrix (CO-PO Mapping) [Food Processing]

COs	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3	PSO-4	Relevant to Local/ National/ Global/ Regional needs
CO-1	2	-	-	-	-	-	-	-	-	-	-	1	1	-	-	-	National
CO-2	2	2	-	-	-	-	-	-	1	-	1	-	-	2	-	-	National
CO-3	3	3	3	3	-	1	-	-	1	1	-	2	2	3	3	3	Global
CO-4	2	2	-	-	-	-	-	-	1	-	1	-	-	2	-	-	National
CO5	2	-	3	2	3	1	-	-	1	2	-	3	2	3	3	3	Global

3 – HIGH, 2 – MEDIUM, 1 - LOW

Open Elective Course for Other Disciplines			
NUTRITION AND HEALTH			
Course Code:	23FT3041	Course Credits:	3
Teaching Hours / Week (L: T: P):	3:0:0	CA Marks:	40
Total Number of Teaching Hours:	45	END-SEM Marks:	60
Focuses on	Employability	Integrates cross-cutting issues	Environment
Course Pre-requisites: 1. Fundamental understanding of biological concepts (such as cell structure, metabolism, digestion, and human anatomy) is essential to grasp the concepts in nutrition and health. 2. Students should be able to understand statistical data. Basic math skills are required for analyzing nutritional data and assessing dietary intake.			
Course Description: a. The course covers key concepts such as nutrition, malnutrition, and the relationship between food, nutrition, and health. b. Students develop an understanding about macronutrients (carbohydrates, proteins, fats), micronutrients (vitamins, minerals), water, and fiber, with emphasis on digestion, absorption, and their roles in health. c. This course provides need base developments in functional foods and their health benefits.			
Course Objectives: This course will enable the students- 1. To provide an understanding of the basic principles of nutrition and its importance for human health. 2. To familiarize students with various nutrients, their roles in the body, and how they contribute to health. 3. To explore the relationship between diet, disease prevention, and health maintenance. 4. To develop an understanding of nutritional needs at different life stages and health conditions 5. To critically analyse food labelling, health claims, and their influence on consumer behaviour.			
Course Outcomes: On completion of the course, learner will be able to- 1. Explain the foundational concepts of nutrition and familiarize with various methods of assessing nutritional status. 2. Summarize the significance of fiber, the food pyramid, and food exchange lists for balanced diet planning. 3. Analyse nutritional requirements across different life stages. 4. Explain the role of nutrition in the prevention and management of various diseases. 5. Analyse various forms of malnutrition, its causes, and the global and national efforts aimed at addressing these issues to improve health outcomes and ensure food security.			
Unit I	Introduction to Nutrition and Health		7 Hours

Definition of nutrition, food as source of nutrients, Functions of food -Physiological, psychological and social, Understanding relationship between food, nutrition and health, Concept of a balanced diet and its role in disease prevention, Methods of assessing nutritional status such as dietary surveys, Anthropometric measurements, Biochemical tests		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning:	
Unit II	Nutrients and Food Groups	12 Hours
Macronutrients, their sources, functions, deficiency/ excess disorders: Carbohydrates, Proteins, Fats, Micronutrients, their sources, functions, deficiency/ excess disorders: Vitamins, Minerals Fiber, their types and significance, Overview of digestion and absorption of nutrients, Food Pyramid, Food Exchange list, Selection, nutritional contribution and changes during cooking of the following food groups: Cereals and Pulses, Fruits and vegetables, Milk & milk products, Eggs, Meat, poultry and fish, Fats and Oils		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning: Case study on animal-based and plant-based sources of protein and their role in the role in muscle repair, enzyme production, and immune function.	
Unit III	Nutrition Across the Lifespan	6 Hours
Nutritional needs during pregnancy and lactation, Infant and child nutrition, Nutrition in adolescence, adulthood, and older age, Nutritional Needs of Athletes and sports person, Sports supplements and their role in performance, Nutrition needs for people working in extreme condition (Army and Navy person)		
Pedagogy	ICT Teaching / Power Point Presentation and Videos	
	Self-study / Do it yourself:	
	Experiential Learning Topics:	
	Case Study / PBL - Project Based Learning: Case study on benefits and potential risks of sports supplements like protein powder, creatine, and BCAAs.	
Unit IV	Nutrition and disease prevention	10 Hours

Nutrition and disease prevention in following
 Obesity -Causes, health risks, and prevention, Role of diet in weight management, cardiovascular diseases-Types, causes and symptoms, Role of nutrition in preventing heart disease and hypertension, Impact of fats, salt, and sugar on heart health, Diabetes- Type 1 vs. Type 2 diabetes, Nutritional strategies for managing diabetes, Cancer Prevention- Impact of nutrition on cancer risk, natural antioxidants and their role in cancer prevention, Relation between Nutrition and Immune health, Key nutrients that support immune health (e.g., Vitamin C, zinc)

Pedagogy	ICT Teaching / Power Point Presentation and Videos
	Self-study / Do it yourself:
	Experiential Learning Topics:
	Case Study / PBL - Project Based Learning: Case study on benefits of consuming fruits, vegetables, and whole grains in cancer prevention.

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Unit V	Global Nutrition Issues and Dietary guidelines	10 Hours
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Malnutrition-
 Forms of malnutrition -undernutrition, overnutrition, Micronutrient deficiencies, Causes and global consequences, National and global efforts to improve nutrition (e.g., fortification, supplementation)

Food Security
 Access to healthy food around the world, Strategies to combat food insecurity and hunger, National and international dietary guidelines (e.g., USDA, WHO), The role of nutrition education in public health, The importance of food labelling and reading food labels

Pedagogy	ICT Teaching / Power Point Presentation and Videos
	Self-study / Do it yourself:
	Experiential Learning Topics:
	Case Study / PBL - Project Based Learning: Case study on misleading labelling of any food product and its consequences

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Learning Resources		
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Text Books:

1. Smolin, Lori A., and Grosvenor, Mary B., Nutrition: Science and Applications, 7th Edition, McGraw-Hill Education, 2023
2. Whitney, Eleanor Noss, and Rolfes, Sharon Rady, Understanding Nutrition, 16th Edition, Cengage Learning, 2022.
3. Gandy, Joan (Ed.), The Oxford Handbook of Nutrition and Dietetics, 3rd Edition, Oxford University Press, 2023.
4. Bamji MS, Krishnaswamy K, Brahman GNV, Textbook of Human Nutrition, 3rd edition. Oxford and IBH Publishing Co. Pvt. Ltd. 2009
5. Srilaxmi, B., Nutrition Science, 7th Edition, New Age International Publishers, 2022

Reference Books:

1. Cummings, Jennifer H., and Allen, Andrew, The Science of Nutrition, 4th Edition, Elsevier, 2022.
2. Wardlaw, G. M., & Hampl, J. S., Contemporary Nutrition: A Functional Approach, 11th Edition, McGraw Hill, 2018.
3. National Institute of Nutrition (India), Dietary Guidelines for Indians: A Manual, ICMR, 2011.
4. Indian Council of Medical Research (ICMR), Nutrient Requirements and Recommended Dietary Allowances for Indians, 2020.

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CO-2	3	2	2	1	2	2	1	1	1	-	-	1	2	1	1	3	Global
CO-3	1	2	2	1	2	2	1	1	1	-	-	1	2	1	1	3	Global
CO-4	3	2	2	2	2	3	1	1	1	1	1	2	2	1	1	3	Global
CO-5	3	2	2	2	2	3	1	2	-	-	3	2	2	1	2	3	Global

3 – HIGH, 2 – MEDIUM, 1 - LOW